

Type 0679H

Square Ceramic Surface Mount Quick Acting Fuse

HF 0679H Series – 2410 Size

RoHS Compliant

Features

- Quick Acting, **2410 SMD**
- Compatible with 260°C, IR Pb-free solder process
- Wide range of current rating from 250mA to 15A
- Wide operating temperature range, **-55°C to 125°C**
- Tape & Reel for auto-insert SMD process
- Full compliance with EU Directive 2011/65/EU and amending directive 2015/863, Compliant with exemption 7(a)
- Halogen Free, (MSL = 1)
- AEC-Q Compliant
- Meets Bel automotive qualification*
- * - Largely based on internal AEC-Q test plan



AEC-Q Compliant

Applications

- Notebook
- LCD monitor
- PC computer
- Office electronic equipment
- Industrial equipment
- Medical equipment
- POE, POE+
- LCD / LED monitor
- Power supply
- LCD / LED TV

HALOGEN FREE = HF

Electrical Characteristics

(UL/CSA/STD. 248-14)

Testing Current	Blow Time	
	Minimum	Maximum
100%	4 Hrs.	N/A
200%	N/A	5 Sec

(250mA and 12.5A follow to GB/T9364.7-2016)

Testing Current	Blow Time	
	Minimum	Maximum
100%	4 Hrs.	N/A
200%	N/A	120 Sec
1000%	N/A	100ms

Safety Agency Approvals

Safety Agency	Safety Agency Certificate	Ampere Rating/ Voltage Rating	Ampere Range / Volt @ I.R. ability*
CULUS	E506667	250mA-3A/350V AC 72V DC 3.5A-15A/350V AC 60V DC	250mA-3A/350V@ 50A AC 72V@100A DC 3.5A-5A/350V@50A AC 60V@150A DC 6.3A-15A/350V@50A-AC 60V@100A DC
TUV	R 50368185 Tested according to IEC 60127-1:2006+A1+A2 IEC 60127-7:2015	250mA-3A/350V AC 72V DC 3.5A-15A/350V AC 60V DC	250mA-3A/350V@ 50A AC 72V@100A DC 3.5A-5A/350V@50A AC 60V@150A DC 6.3A-15A/350V@50A-AC 60V@100A DC
CQC	CQC25012495846	250mA, 12.5A/100V AC	250mA, / 100V@35A AC 12.5A/ 100V@125A AC

*I.R.= Interrupting Rating = Short Circuit Rating(Amps)

Physical Specifications

Materials	Body : Ceramic Terminations : Silver Plated Caps On Fuse :
Marking	"Current Rating", "Q", "H"—laser marked on ceramic tube, "bel" stamped in end caps. On Label : "bel", "0679H", "Current Rating", "Voltage Rating", "Interrupting Rating", "Appropriate Safety Logos" and "RoHS" (China RoHS compliant).

Specifications subject to change without notice

Environmental Specifications

Shock Resistance	MIL-STD-202G, Method 213B, Test Condition 1 (100 G's peak for 6 milliseconds; Sawtooth waveform)	High temperature storage	MIL-STD-202 Method 108
Vibration Resistance	MIL-STD-202G, Method 201A (10-55 Hz, 0.06 inch, total excursion).	Temperature cycling	JESD22 Method JA-104, Test Condition B
Salt Spray Resistance	MIL-STD-202G, Method 101E, Test Condition B (48 hrs.).	Biased humidity	MIL-STD-202 Method 103, 85°C/85% RH with 10% operating power for 1000 hrs.
Insulation Resistance	MIL-STD-202G, Method 302, Test Condition A (After Opening) 10,000 ohms minimum.	Operational life	MIL-STD-202 Method 108, Test Condition D
Solderability	MIL-STD-202G, Method 208H	Resistance to solvents	MIL-STD-202 Method 215
Resistance to solder Heat	MIL-STD-202G, Method 210F, Test Condition C. Top Side (260°C, 20 sec) MIL-STD-202G, Method 210F, Test Condition D. Bottom Side (260°C, 10 sec)	Mechanical shock	MIL-STD-202 Method 213, Test Condition C
Thermal Shock	MIL-STD-202G, Method 107G, Test Condition B (-65°C to +125°C).	Vibration	MIL-STD-202 Method 204
Operating Temperature	-55°C to +125°C	Resistance to soldering heat	MIL-STD-202 Method 210, Test condition B
Moisture Sensitivity Level	1 (According to IPC J-Std-020)	Thermal shock	MIL-STD-202 Method 107
		Solderability	J-STD-002
		Board flex(SMD)	AEC-Q200-005
		Terminal strength	AEC-Q200-006
		Electrical characterization	3 temperature electrical

Electrical Specifications

Part Number	Ampere Rating (A)	Typical Cold Resistance (ohms)	Volt-drop @100% In (Volt) max.	Voltage and Interrupting Ratings	Melting I ² T <10m Sec (A ² Sec)	Melting I ² T @10 In (A ² Sec)	Maximum Power Dissipation (W)	Agency Approvals		
										
0679H0250-XX	250mA	0.55	0.530	See Table of Safety Approvals on Page 1 for Voltage and associated Interrupting Ratings	0.01	0.02	0.13	Y	Y	Y
0679H0375-XX	375mA	0.32	0.480		0.04	0.04	0.18	Y	Y	
0679H0500-XX	500mA	0.22	0.470		0.08	0.08	0.24	Y	Y	
0679H0630-XX	630mA	0.17	0.410		0.15	0.15	0.26	Y	Y	
0679H0750-XX	750mA	0.14	0.380		0.24	0.26	0.29	Y	Y	
0679H1000-XX	1A	0.09	0.280		0.51	0.54	0.28	Y	Y	
0679H1250-XX	1.25A	0.068	0.250		0.21	0.22	0.31	Y	Y	
0679H1500-XX	1.5A	0.053	0.250		0.32	0.29	0.38	Y	Y	
0679H2000-XX	2A	0.035	0.240		0.62	0.68	0.48	Y	Y	
0679H2500-XX	2.5A	0.028	0.240		0.96	1.13	0.60	Y	Y	
0679H3000-XX	3A	0.022	0.220		1.6	1.8	0.66	Y	Y	
0679H3500-XX	3.5A	0.019	0.220		2.0	2.2	0.77	Y	Y	
0679H4000-XX	4A	0.018	0.220		3.1	3.5	0.88	Y	Y	
0679H5000-XX	5A	0.014	0.200		5.3	5.5	1.00	Y	Y	
0679H6300-XX	6.3A	0.011	0.190		8.7	8.3	1.20	Y	Y	
0679H7000-XX	7A	0.010	0.175		11.1	10.8	1.23	Y	Y	
0679H8000-XX	8A	0.0085	0.170		14.8	14.1	1.36	Y	Y	
0679H9100-XX	10A	0.0064	0.150		25.7	25.7	1.50	Y	Y	
0679H9120-XX	12A	0.0054	0.140		41.0	38.9	1.68	Y	Y	
0679H9125-XX	12.5A	0.0052	0.170		41.0	42.1	2.13	Y		Y
0679H9150-XX	15A	0.0038	0.130		76.7	103.5	1.95	Y	Y	

Consult manufacturer for other ratings

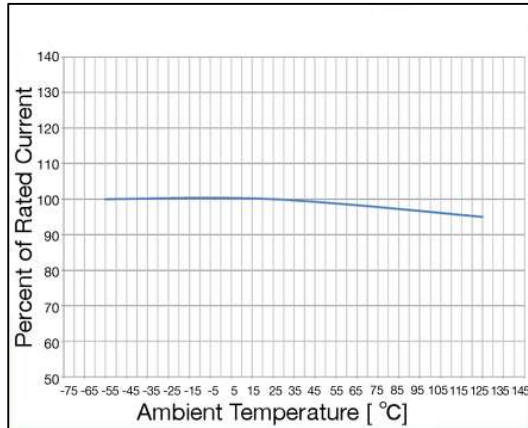
XX - Packaging code (see "ordering information")

NOTES:

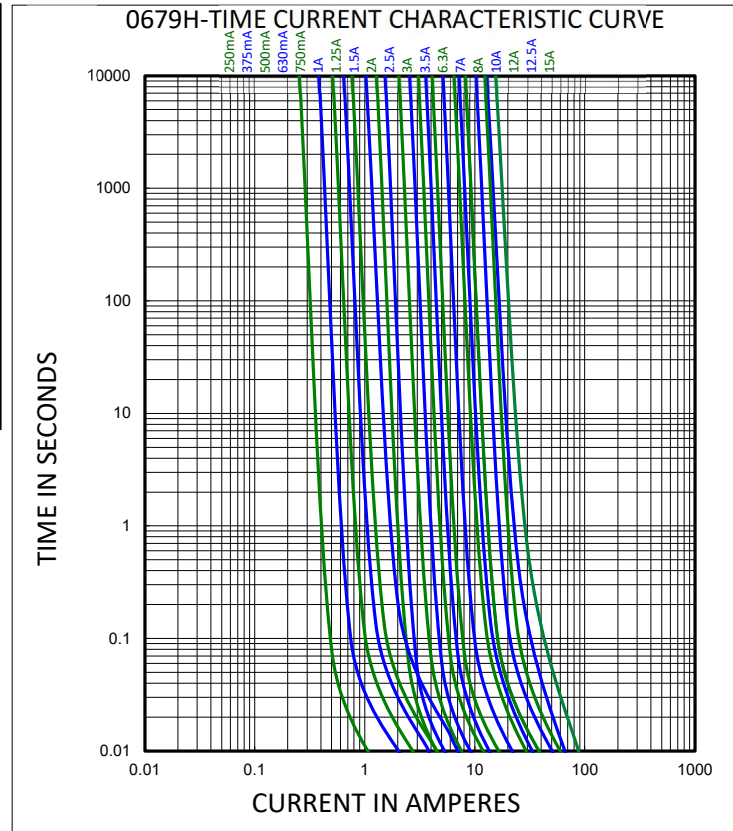
All tests were conducted with the fuses soldered to a printed circuit boards with a nominal thickness of 1.6 mm. The copper test circuit trace was a printed circuit with an overall length of 100 mm, copper thickness/width as described below. The printed circuit boards were mounted by screws to a test fixture having brass blocks for connection of the test leads. All samples were soldered to the test boards by the manufacturer.

Fuse rating	Test Board Trace Dimensions
250mA-5A	1 oz. copper, 5.0mm wide.
6.3A-15A	3 oz. copper, 10mm wide.

Temperature Derating Curve

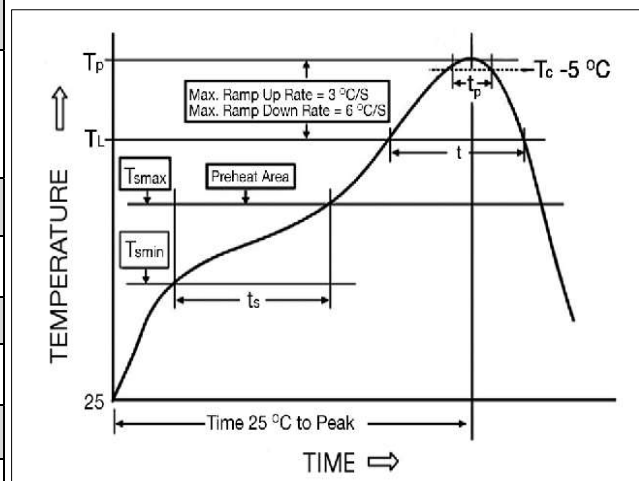


Average Time Current Curve



Soldering Parameters

IR Reflow Profile (IPC/JEDEC J-STD-020D)	
Preheat & Soak	
Temperature min (T_{smin})	150 °C
Temperature max (T_{smax})	200 °C
Time (T_{smin} to T_{smax}) (t_s)	60-120 seconds
Average ramp-up rate (T_{smax} to T_p)	3 °C/second max.
Liquidous temperature (T_L)	217 °C
Time at liquidous (t_L)	60-150 seconds
Peak temperature (T_p)	260 °C max
Time (t_p) within 5 °C of the specified classification temperature (T_c)	30 seconds
Average ramp-down rate (T_p to T_{smax})	6 °C/second max.
Time 25 °C to peak temperature	8 minutes max.

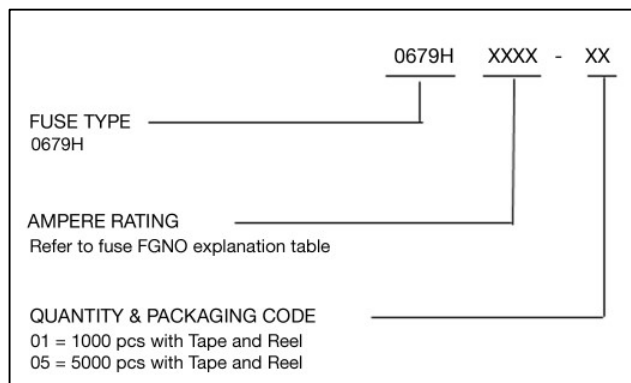


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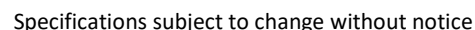
0679H=0679H; [XXXX]=Ampere Rating; XX=See Ordering Information as below

Fraction	Decimal	Amps	Bel FGNO[XXXX]
1-1/4	1.25	1.25	1250
1-1/2	1.50	1.5	1500
	2.0	2	2000
2-1/2	2.5	2.5	2500
	3.0	3	3000
3-1/2	3.5	3.5	3500
	4.0	4	4000
	5.0	5	5000
	6.3	6.3	6300
	7.0	7	7000
	8.0	8	8000
		10	9100
		12	9120
		12.5	9125
		15	9150

Ordering Information



Packaging Tape & Reel	Packaging Specification	Quantity	Quantity & Packaging Code
12 mm wide tape with 13 inches Diameter reel	EIA Standard 481-E	5000	05
12 mm wide tape with 7 inches Diameter reel	EIA Standard 481-E	1000	01



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